MODULE 15 / LESSON 01

RAG Supplies Evidence at Request Time

Retrieval-augmented generation (RAG) finds relevant evidence and includes it in the model context before generation.

DSACADEMY

CORE CONCEPT

Parsing Recovers Content from Source Files

Document parsing converts a source format into structured text and metadata. A parser may need to recover:

- **headings and paragraphs;**
- **tables;**
- **page numbers;**
- **lists and reading order;**
- **captions;**
- **document identifiers.**

RUN IT AND INSPECT THE RESULT

Cleaning Must Preserve Provenance

```
chunk.metadata = {
    "source_id": "policy-2026-v3",
    "page": 7,
    "section": "Refunds",
}
```

OUTPUT

Each future retrieval result can identify its exact source location.

Removing exact repeated headers is reasonable. Removing every line containing Policy may delete real content. Record the cleaning rule and compare before/after counts.

REASONING SEQUENCE

Chunking Defines the Unit of Retrieval

Frame

whether a complete idea stays together;

1

Transform

embedding specificity;

2

Validate

retrieval recall;

3

Explain

prompt size;

4

A 900-token section becomes: The second keeps boundary context but may retrieve repeated statements.

CORE CONCEPT

Embeddings Place Chunks and Queries in One Vector Space

An embedding model maps text into a fixed-length vector. For vector retrieval: Changing embedding model usually requires rebuilding the stored index.

- **embed every document chunk;**
- **store chunk vector plus text and metadata;**
- **embed the user query with the same compatible model;**
- **compare the query vector with chunk vectors;**
- **return the highest-ranked allowed chunks.**

RUN IT AND INSPECT THE RESULT

Cosine Similarity Compares Vector Direction

```
import numpy as np

a = np.array([1.0, 0.0])
b = np.array([1.0, 1.0])
similarity = a @ b / (np.linalg.norm(a) * np.linalg.norm(b))
print(round(similarity, 3))
```

OUTPUT

```
0.707
```

Dot product = 1 ||a|| = 1 ||b|| = sqrt(2) Cosine = 1/sqrt(2) ≈ 0.707

CORE CONCEPT

A Vector Store Connects Vectors to Evidence

A vector store keeps: An exact search compares all vectors. Approximate indexes trade some recall for lower latency at scale. Apply authorization filters before or during retrieval. Filtering after retrieval can leak whether protected documents exist and waste candidate slots.

embedding vector;

chunk text or reference;

source metadata;

access metadata;

stable chunk ID.

REASONING SEQUENCE

Retrieval Selects Top-k Evidence

Frame

query;

1

Transform

returned text;

2

Validate

scores and score direction;

3

Explain

source diversity;

4

Relevant policy appears at rank 4. With k=3, it is missed. With k=5, it is included alongside two irrelevant chunks. The choice must be evaluated across a query dataset, not one example.

RUN IT AND INSPECT THE RESULT

Context Assembly Builds the Evidence Packet

```
context = "\n\n".join(
    f"[{doc.metadata['source_id']} p.{doc.metadata['page']}]\n{doc.page_content}"
    for doc in selected_docs
)
```

OUTPUT

A labelled context packet whose citations can map back to retrieved documents.

After deduplication, four chunks consume 2,500. The assembler selects the highest-value set under 2,000 while retaining at least one chunk from each required source.

QUALITY GATE

Grounded Answers Need Verifiable Citations

- [x] cited source was retrieved;
- [x] cited passage supports the nearby claim;
- [x] no unsupported claims were added;
- [x] quotation and numbers match;

RISK REVIEW

A RAG Failure Has an Owning Stage

01 parsing lost text;

02 cleaning changed meaning;

03 chunk boundary split evidence;

GUIDED LAB

Guided Lab: Build RAG from Scratch

01

parse and visually verify source pages

02

preserve source, page, section, and access metadata

03

compare two chunking strategies

04

embed chunks and queries with one model